

A mixed TCC–Born inverse of LEEM Fourier optics: algorithm selection

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26 August 2026

Abstract

We invert a two-dimensional through-focal LEEM intensity stack for the exit wave $\psi_0 = \sigma e^{i\phi}$ against the bilinear Fourier-optics (FO) kernel of Yu, Lau and Altman [1]. Among discrete inverses that are identities of that kernel, the cheapest large-phase estimator at modelled $N = 128$ is a mixed transmission-cross-coefficient (TCC)–Born homotopy: vacuum 2×2 Tikhonov initialization, rank- M TCC (or exact rank-1 under perfect coherence) apply, remainder-corrected Born updates on the same Fourier-diagonal 2×2 , per-bin skip $t \in \{0, 1\}$, and optional algebraic Armijo on the exact quartic line energy. This note records *every* candidate examined against that mix, with a Pick / Component / Not-pick verdict and the Lean witness or an explicit informal tag. The scientific inverse statement is the companion note [5]; algebraic claims are machine-checked in Lean 4.

1 Problem and selection criteria

The forward map is bilinear FO intensity [1]

$$I(\mathbf{r}, \Delta z) = \iint \Psi(\mathbf{q}) \Psi^*(\mathbf{q}') R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) e^{2\pi i(\mathbf{q}-\mathbf{q}') \cdot \mathbf{r}} d^2 q d^2 q', \quad (1)$$

with $\Psi = \mathcal{F}[\psi_0]$ and kernel support $|\mathbf{q}|, |\mathbf{q}'| \leq q_{\text{ap}}$. On a finite frequency group G and a finite defocus index κ , the same bilinear form is `ihat` on the 2D kernel `R_F02` [4, 5].

A candidate is a **Pick** for 2D large ϕ only if

1. it is an *identity* of `ihat/R_F02` (FO-faithful), and
2. it is strictly cheaper than a dense pair apply KN^2 at $N = 128$ in the cost model `dftCost(N) := N \log_2 N` (`dftCost` is a definition, not an FFT existence theorem).

A method may be a **Component** of the winner without being a standalone 2D inverse. **Not pick** means it fails faithfulness, cost, or the experiment geometry (through-focal LEEM with known aberration χ). Convergence rates, statistical noise models and Banach fixed-point theorems are not FO identities and are tagged *informal*.

2 Winner mix

The locked 2D large-phase estimator is

1. **Init.** Vacuum 2×2 Tikhonov on R_{FO} in 2D (`stage1Pair2/tikhonovXhat2`), default $K = 2$ independent defoci.
2. **Apply.** Rank- M TCC (`ihat_tcc`) or exact rank-1 autocorrelation under **PerfectCoherence**. Rank policy `recommendTccRank`: $M = 1$ if perfectly coherent or if using the quartic line search, else $M \leq 8$ at $N = 128$.

3. **Update.** Remainder-corrected Born (`mixedBinStep`) on the same Fourier-diagonal 2×2 .
4. **Skip.** Per-bin `remainderWeight` $\in \{0, 1\}$ (`mixed2DMix`): quiet $t = 0$ keeps stage 1; loud $t = 1$ takes the full FO remainder.
5. **Line search (optional).** Algebraic Armijo (`lineDescentStep`) on stack fidelity `lineFid`.
6. **Partner.** Hermitian data and odd $qmap$ recover $\delta(-\xi) = \overline{v(\xi)}$ (`mixed2D_conj_partner`).

FO-faithfulness is the exact quadratic homotopy (`ihat_homotopy_cubic_zero`): there is no cubic remainder. Cost: a rank- $M \leq 8$ Born step is cheaper than dense at $N = 128$ (`bornStepCost_lt_dense_128`, `hybridCost_lt_dense_succ`). The one-dimensional sinusoid of Yu *et al.* remains Jacobi–Anger (`sinusoidJ`), not a branch of `mixed2D`.

Quiet skip is the linearized / weak-object branch (`mixedSpectrumMix_skip_step`); there is no separate weak-phase inverse. `mixed2D` with free schedules (t, damp) stays the general pair API; `mixed2DMix` is the recommended skip policy.

3 Picked components

Vacuum 2×2 Tikhonov / Schiske–Wiener. **Pick as init** and as the quiet-bin map. For $\alpha > 0$, `tikhonovXhat2` is the unique minimizer of the bin energy `tikhonovJ2` (`tikhonov2_unique`). One Gauss–Newton step from vacuum *is* that solve (`vacuumGN_eq_stage1Pair2`). Default $K = 2$: $K = 1$ always has a kernel on $(X(\mathbf{q}), \overline{X(-\mathbf{q})})$ (`exists_one_defocus_pair_kernel`). Closed-form Cramer, not an iterative 2×2 solver.

Rank- M TCC apply. **Pick as apply.** Hopkins expansion `ihat_tcc` with modelled cost $KM \cdot 2 \cdot N \log_2 N$. At $N = 128$, $M \leq 8$ is cheaper than dense (`tccApplyCost_lt_dense_128`); $M = 9$ is still cheaper (`tccApplyCost_lt_dense_128_nine`); $M \geq 10$ is not. The Born cap stays $M \leq 8$. Zero-weight padding of unused modes is free (`tccKernel_insert_weight_zero`).

Perfect-coherence rank-1 autocorrelation. **Pick when PerfectCoherence** ($q_{\text{ill}} = \Delta E = 0$): `ihat_R_FO_of_perfect` / `ihat_rank1`. Not picked from “envelope ≈ 1 ”: `exists_R_FO_ne_R0` on a column with $\Delta E = 0$ but $q_{\text{ill}} \neq 0$.

Remainder-corrected Born / quadratic homotopy. **Pick as update.** The true ray is $\hat{I}(\text{vac} + t\delta) = \hat{I}(\text{vac}) + tDI + t^2\hat{I}(\delta)$. `bornModel` interpolates stage-1 ($t = 0$) and full FO ($t = 1$). Loud bins use $t = 1$; quiet bins use $t = 0$.

Per-bin skip remainderWeight. **Pick as skip.** The control is a bit (`remainderWeight_eq_zero_or_one`). Global t and fractional $t = \min(1, \|\hat{I}\|/\eta)$ are not the mix ([Section 4](#)).

Algebraic Armijo on the exact quartic. **Pick as optional 1D mix** after a Born direction. Along $x_0 + s \cdot d$ the residual is degree 2, so stack fidelity `lineFid` is a real quartic (`lineFid_quadEnergy`). `descentStep` decreases energy when the cubic slope at 0 is negative (`lineFid_descent`). Full NLS on the ray is the companion `nlsJ_line`; the FO 1D mix stays fidelity-only because bins already use α in `mixedBinStep`.

Hermitian partner / odd qmap. **Pick as recovery.** Hypotheses are $q(-\xi) = -q(\xi)$ and Hermitian data, not odd $|G|$.

Jacobi–Anger for a 1D sinusoid. Pick as a 1D special case (sinusoidJ, stage2Sinusoid). **Not pick as a 2D inverse:** the Bessel ladder has M modes, bilinear FO has M^2 pairs (exists_modeSet_lt_modePairs).

LR+D apply. Optional component, not the default apply. Tighter truncation bound than plain TCC, strictly costlier (tccApplyCost_lt_lrPlusDiagApplyCost); still cheaper than dense at the Born cap.

4 Not pick: FO-map mismatch

Transport of intensity / Fresnel multiplier. Wrong PDE; the multiplier is nonzero outside the aperture where FO is blind (exists_interior_R_F0_ne_tie, exists_interior_R_F02_ne_tie, R_F0_ne_tie_outside).

Gerchberg–Saxton / HIO / error reduction / R_{CTF} . The coherent intensity is $R_{CTF} = R_{FO} \Gamma_C \Gamma_S$; $\Gamma_S \neq 1$ off axis (exists_R_CTF_ne_R_F0). HIO/ER use the same misspecified projector (exists_hio_kernel_ne_R_F0).

Coherent projector R_0 as FO. Partial spatial coherence: exists_R_F0_ne_R0.

Single-defocus Wiener. The vacuum 2×2 always has a kernel (exists_one_defocus_pair_kernel).

Weak-phase CTF $\sin(2\pi\chi_S)$ as the map. Can vanish while the FO slice does not (exists_weakPhase_sin_zero_R_F0_ne_zero). Quiet Mix skip is the linearized path; this sine is not that path.

Radial / $\|\mathbf{q}\|$ -only 2D kernel. E_S contains $\mathbf{q} \cdot \mathbf{q}'$ (exists_R_F02_ne_R_F0_of_norms, exists_aS2_ne_radial_aS). Cartesian R_{F02} stays the 2D kernel.

Jacobi–Anger as a 2D inverse / Bool dispatcher on mixed2D. exists_modeSet_lt_modePairs. Types differ: scalar ϕ -cost versus Fourier-bin 2×2 .

First-order Rytov increment. $i\phi$ is not $e^{i\phi} - 1$ on FO (exists_rytov_inc_ne_fo_phase).

Forgetful Kaczmarz. Last-row only; not the tikhonovJ2 minimizer for $K \geq 2$ (exists_kaczmarzSweep_ne_tikhonovXhat2) and costlier than one Cramer solve (binSolveCost_lt_kaczmarzCost_one_sweep).

Unit Gauss–Newton as a line critical point. The exact cubic at a unit GN step ($B = -A$) is not identically zero (lineCubic_exactGN); a unit step can raise energy (exists_unit_step_energy_increase).

Gauss–Newton Hessian as the quartic Hessian. Drops the remainder coupling $2\text{Re}(\overline{AC})$ (exists_quadA2_ne_gn).

Quadratic Newton candidate $-A_1/(2A_2)$. Ignores A_3, A_4 ; not a cubic root; can raise energy (exists_newtonCandidate_not_critical, exists_newtonCandidate_energy_increase).

Global or fractional homotopy t as skip. Exact FO is loud $t = 1$; stage-1 is quiet $t = 0$. Distinct t disagree on occupied remainder (`exists_bornModel_t_disagree`); loud $t = 0$ drops the remainder (`bornRhs_t0_ignores_loud`). Fractional $t = \min(1, \|\hat{I}\|/\eta)$ is neither endpoint.

5 Not pick: faithful or equivalent, but not the cheap mix

Dense / generic-background Gauss–Newton. FO-faithful (exact quadratic remainder) but `ihatJac` off vacuum is not Fourier-diagonal and is KN^2 , not $O(KN \log N)$.

Real-space Born–GN as the cheap mix. Same obstruction as dense GN.

Always-on quartic line search. Faithful 1D mix, but `recommendTccCap` then forces $M = 1$. Rank-1 line search beats dense at $N = 128$ (`lineSearchCost_lt_dense_128_rank1`); rank-8 line search does not. Born $M \leq 8$ is the cheaper 2D default; Armijo stays optional.

Tikhonov line replacing `lineFid`. `nlsJ = lineFid +  $\alpha \sum \|x_0 + sd\|^2$  (nlsJ_line).` Bins already use α . Fidelity-only `lineFid` can be flat while $\|x_0 + sd\|$ is unbounded (`exists_lineFid_flat_unbounded`).

Two-source special case in `recommendTccRank`. `ihat_twoSource` is `ihat_tcc` at $M = 2$.

$\Delta E = 0$ as rank-1 apply. Chromatic envelope 1 is not `PerfectCoherence` (`exists_R_F0_ne_R0` on `nacPC`).

Landweber / CGLS / CG as bin solvers. Dimension 2: exact CG recovers Cramer after the same $O(K)$ Gram work; rates are informal. Stage-1 stays closed-form `tikhonovXhat2`.

Recursive least squares / Kalman. Proper accumulation of the Gram is the same estimator as batch Cramer, not cheaper. Forgetful streaming is Kaczmarz (already rejected).

Three-point $\{0, \text{descentStep}, 1\}$ min. Same `lineSearchCost` as Armijo; not required (T15 already has descent). Fixed $\{0, \frac{1}{2}, 1\}$ does not guarantee descent.

LR+D as the default apply. See [Section 3](#): optional tighter bound, not cheaper.

Near-perfect coherence as hard $M = 1$. Rank-1 apply needs `PerfectCoherence`. Near-PC belongs under `ihat_tcc_trunc_bound` / `chooseTccRank`, not a hard cap of 1.

Formal $K > 2$ always better / iterate count $T = 1$ because quadratic. $K = 2$ is the algebraic default; extra K is SNR / common CTF zeros (informal). Quadratic FO is the *model*, not one-step exactness of Born; T stays API (`hybridCost` is parametric in all T).

6 Not pick: not an FO identity, or a different experiment

Each item below was examined and left informal or out of geometry. No new `exists_*` is claimed.

Cardano roots of `lineCubic`. Stationarity is a cubic; roots are not constructed. Armijo is the encoded 1D mix.

Trust-region / dogleg / cubic regularization. The quadratic model is not the FO quartic; a cubic model returns to Cardano.

Complex-step line search. A complex step is sesquilinear in $(\sigma, \bar{\sigma})$, not `quadPoly`. Real $s \in \mathbb{R}$ only.

Nesterov / FISTA / Polyak / Heavy-ball. Acceleration rates, not FO identities; same bucket as GN/Born convergence.

L-BFGS / BFGS / Levenberg–Marquardt. Secant / damped-GN models inherit the GN Hessian gap (`exists_quadA2_ne_gn`) and lose the Fourier-diagonal 2×2 Born loop.

Anderson / DIIS mixing. Multi-history residual LS, not the scalar damp of `mixedBinStep`; not cheaper per outer step.

Wirtinger / ADMM / PhaseLift convergence. Not FO identities. The PC kernel mismatch is already `exists_R_F0_ne_R0`.

Banach FPT / Picard iterate convergence. `picard_unique_of_lip` is only a Lipschitz implication. Existence and rates are not encoded.

TV / Split-Bregman / ℓ^1 . Drops the Cramer uniqueness story of $\alpha \|\cdot\|^2$. FO-faithfulness of the mix comes from the quadratic homotopy, not the penalty.

Weighted LS / VST / $1/|h|^2$. Statistical; `tikhonovJ2/nlsJ` are unweighted ℓ^2 . No probability space.

Huber / truncated least squares. A robust $\rho(r)$ breaks $\|\text{quadPoly}\|^2 \rightarrow \text{quartic}$, so it is not the FO 1D mix.

Richardson–Lucy / Poisson EM. No probability space; not bilinear FO.

MAL / phase diversity as the large- ϕ inverse. GN stacked over extra defoci when χ is *unknown* [3]. The mix assumes known χ in `R_F02` and is cheaper.

Ptychography / Wigner-distribution deconvolution. Probe–diffraction geometry, not a through-focal LEEM stack.

Multi-slice / dynamical scattering. A volume propagator, not bilinear `ihat`.

Learned / unrolled neural inverse. Not an FO identity.

Image alignment / drift as the inverse. Preprocessing. Inverse inputs are already Fourier bins. Alignment and the current-to-defocus map remain informal in the inverse note.

Contrast-normalized $y/\langle I \rangle$ or $y_k - y_0$. Experiment convention. The mix uses raw y and vacuum-subtracted `yLin`.

Defocus interpolant on Δz . κ is given; missing frames shrink the index. Uneven Δz is already an arbitrary map $\kappa \rightarrow \mathbb{R}$.

Energy-filtered / spectroscopic stack as κ . The inverse index is defocus, not energy. Bibliographic energy-filtered LEEM is not this mix.

SPLEEM / magnetic contrast. Scalar `objectWave` $\sigma e^{i\phi}$; spin is a different channel.

Pin $\text{Im}(\delta 0) = 0$ for uniqueness. `ihat_gauge`: global phase leaves intensity and $\sum \|\Psi\|^2$ invariant. Gauge theatre.

Deprecate `mixed2D`; restrict to `modeSet`; DC-bin special case. `mixed2D` stays the general pair API. `modeSet` is the 1D JA ladder; outside-disk bins already vanish (T4). DC uses `yLin = y - R_{FO}(0, 0)`; partner at 0 is $v = \bar{u}$.

7 Verdict table

Algorithm	Verdict	Why
Vacuum 2×2 Tikhonov	Component (init)	Unique Cramer; $K = 2$ default
Rank- M TCC	Component (apply)	Cheaper than dense at $N = 128$, $M \leq 8$
PC rank-1 autocorrelation	Component	Exact under PerfectCoherence
Remainder-corrected Born	Component (update)	Exact quadratic FO
<code>remainderWeight</code> $\{0, 1\}$	Component (skip)	Quiet=stage-1, loud=full FO
Quartic Armijo	Optional 1D mix	Exact <code>lineFid</code> ; not always-on
Odd <code>qmap</code> partner	Component	Hermitian recovery
Odd $ G $ as partner hyp.	Not pick	Partner uses $q(-\xi) = -q(\xi)$, not odd cardinality
Jacobi–Anger	1D only	Not a 2D inverse
LR+D	Optional apply	Tighter, not cheaper than TCC
Mixed TCC–Born	Pick (2D large ϕ)	Faithful and cheapest in tree
TIE / Fresnel	Not pick	Wrong PDE; <code>exists_*_ne_tie</code>
GS / HIO / ER / R_{CTF}	Not pick	$\Gamma_S \neq 1$; <code>exists_R_CTF_ne_R_F0</code>
R_0 as FO	Not pick	<code>exists_R_F0_ne_R0</code>
Single-defocus Wiener	Not pick	Vacuum 2×2 kernel
Weak-phase sine as map	Not pick	Zeros inside aperture
Radial 1D-as-2D	Not pick	$\mathbf{q} \cdot \mathbf{q}'$ in E_S
JA as 2D / Bool dispatcher	Not pick	Ladder M vs pairs M^2
Rytov increment	Not pick	$i\phi \neq e^{i\phi} - 1$
Forgetful Kaczmarz	Not pick	Not <code>tikhonovXhat2</code> ; costlier
Unit GN / GN Hessian / Newton A_1, A_2	Not pick	Not the FO quartic
Global / fractional t skip	Not pick	Not endpoints of Born/FO
Dense / real-space GN	Not pick (as cheap mix)	Faithful, KN^2
Always-on line search	Not pick (as 2D default)	Forces $M = 1$
Tikhonov line / 3-point min	Not pick	Companion or redundant
Two-source rank case	Not pick (extra)	Already TCC $M = 2$
$\Delta E = 0$ rank-1	Not pick	Not PerfectCoherence

Algorithm	Verdict	Why
Landweber / CGLS / RLS / Kalman	Not pick	Cramer exact; forgetful=Kaczmarz
LR+D as default	Not pick	Costlier than <code>ihat_tcc</code>
Near-PC as $M = 1$	Not pick	Needs <code>PerfectCoherence</code>
Formal K -count / iterate-count T	Not pick (cost)	$K = 2$ default; T stays API (<code>hybridCost</code>)
Cardano / TR / dogleg / complex s	Informal	Not the encoded 1D mix
Nesterov / FISTA / L-BFGS / LM / Anderson	Informal	Rates / wrong model
Wirtinger / ADMM / PhaseLift conv.	Informal	Not an identity
Banach FPT / Picard conv.	Informal	Only Lip uniqueness
TV / WLS / Huber / RL	Informal	Penalty / noise / Poisson
MAL as large- ϕ inverse	Informal	Unknown- χ only
Ptycho / WDD / multi-slice / SPLEEM / EF- κ	Informal	Different experiment
Learned inverse	Informal	Not an identity
Alignment / I -norm / Δz interpolant	Informal	Preprocessing / convention
Pin $\text{Im}(\delta 0)$	Informal	<code>ihat_gauge</code>
Deprecate <code>mixed2D</code> / restrict <code>modeSet</code> / DC special case	Informal	Keep pair API; JA is 1D; DC is $y - R_{\text{FO}}(0, 0)$
FFT existence / noise models / <code>Approx.</code> truncations	Informal	Not claimed; see Section 9

8 Lean encoding map

Modules: `Forward/Kernel2`, `Inverse/Tikhonov`, `LinearInverse`, `Pipeline`, `LowRank`, `Homotopy`, `LineSearch`, `Plane2`, `Mix`. Theorem list T10–T17 in the inverse statement file of the repository. Sound proofs depend only on `[propext, Classical.choice, Quot.sound]`. No sorry.

9 Not claimed

FFT existence or wall-clock $O(N \log N)$; array DFT = continuum FT; Gaussian/Poisson/MSE/Cramér–Rao; frozen-aperture and first-order tilt truncations already flagged `Approx.` in the forward model; local quadratic convergence of GN/Born; “fastest among all conceivable inverses” (only DFT-based / dense-bilinear / TCC comparisons are proved); uniqueness of the nonlinear inverse (`ihat_gauge`).

References

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